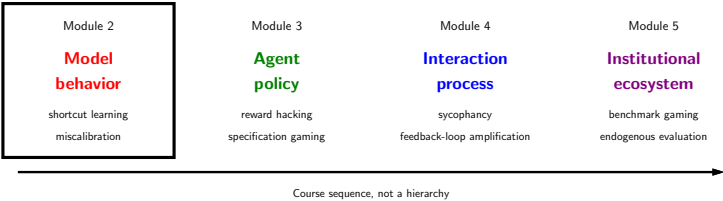




Lecture 4: Learning, Generalization, and Underspecification

CS120: Introduction to AI Safety - October 1, 2026, Zachary Robertson

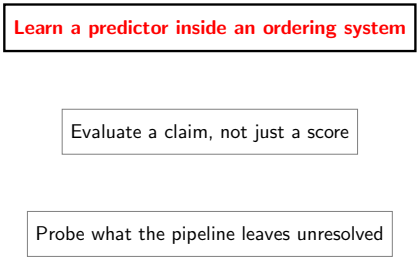
From system assurance to model behavior



How is a predictor's behavior shaped by its development system?
How can that behavior contribute to harm scenarios in the system?



Roadmap



Reflex-based prediction



A predictor function f takes an input x and produces an output y .
Binary classification: $f(x) \in \{-1, +1\}$.

The space of possible predictors $\mathcal{H}$ is the hypothesis class.

Adapted from CS221 Spring 2026, Reflex-based models

Formulate the preference problem

Fix one customer. If the requested item is unavailable, compare substitutes (x, z) .

$x = \text{Apple}$	$z = \text{Spinach}$
$y = -1$	$y = +1$
Selection of x	Selection of z

Intended construct: preference between food types at comparable freshness.

Recorded comparisons may instead confound food type with freshness.

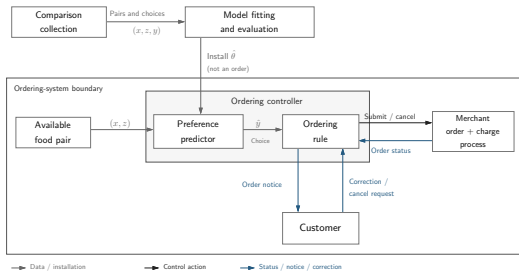
STPA 1: purpose, harm, and hazard

Hypothetical service: equal prices, comparable acceptable freshness, delegated ordering without confirmation of each purchase.

Harm L_F	The customer is charged for the alternative they would reject in favor of the other.
Hazard H_F	A predicted choice can become a nonrefundable purchase without an effective opportunity for the customer to confirm or correct it.
Constraint SC_F	Prevent nonrefundable purchase of an order that would contradict the customer's intended choice.

The hazard is an upstream condition rather than the charge itself.

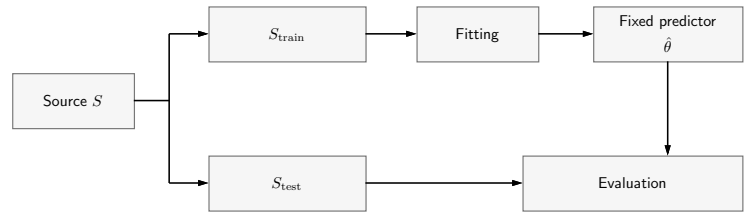
STPA 2: controller, process, and feedback



Submission and commitment can be separated by a delay; commitment makes the charge nonrefundable.

Keep learning and testing separate

Assume independent and identically distributed examples $(x, z, y) \sim S$.



No test-to-fitting or test-to-selection arrow in this design.

Modeling: scores become preference probabilities

Represent alternatives in a common feature space: $x, z \in \mathbb{R}^k$.

Parameters $\theta \in \mathbb{R}^k$ give scores $\theta^\top x$ and $\theta^\top z$.

$$\Pr_\theta(x \succ z \mid x, z) = \frac{e^{\theta^\top x}}{e^{\theta^\top x} + e^{\theta^\top z}}$$

The probability of choosing z is the complement.

Bradley-Terry model; a chosen hypothesis class, not an observed customer law.

Learning: minimize average log loss

$$\ell_{\log}(\theta; x, z, y) = -\log \Pr_\theta(y \mid x, z)$$

$$\hat{R}_{\text{train}}^{\log}(\theta) = \frac{1}{|S_{\text{train}}|} \sum_{(x, z, y) \in S_{\text{train}}} \ell_{\log}(\theta; x, z, y)$$

Learning seeks parameters that minimize this average loss.

With the i.i.d. likelihood: equivalent to maximizing the training likelihood.

Gradient descent is one fitting algorithm.

Inference predicts; the ordering rule acts

Fix the fitted model $\hat{\theta}$. Choose the higher-scoring option:

$$\hat{\theta}^\top x \geq \hat{\theta}^\top z$$

$$\hat{y} = -1: \text{choose } x$$

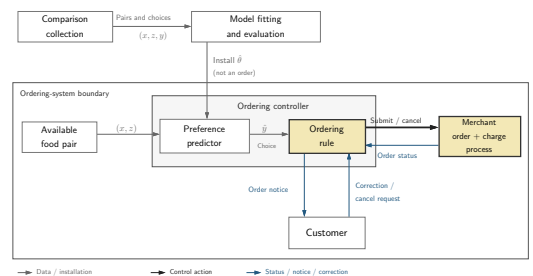
Otherwise

$$\hat{y} = +1: \text{choose } z$$

Ties choose x .

This is a prediction; the ordering rule separately turns it into a purchase instruction.

STPA 3: an unsafe ordering action



Unsafe ordering action: The ordering controller submits x when the intended choice is z , while the order can become nonrefundable before correction.



Roadmap

Learn a predictor inside an ordering system

Evaluate a claim, not just a score

Probe what the pipeline leaves unresolved

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Training, test, and population risk

Switch to bounded zero-one evaluation loss, not fitting log loss.

$$\ell_{01}(\hat{y}_{\theta}(x, z), y) = 1\{\hat{y}_{\theta}(x, z) \neq y\}$$

$\hat{R}_{\text{train}}(\hat{\theta})$ Average error on training examples

$\hat{R}_{\text{test}}(\hat{\theta})$ Average error on held-out test examples

$R_S(\hat{\theta})$ Expected error on the source distribution

$$R_S(\theta) = \mathbb{E}_{(x, z, y) \sim S}[\ell_{01}(\hat{y}_{\theta}(x, z), y)]$$

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What does held-out evaluation support?

Sampling	Test examples are i.i.d. from S .
Independence	The predictor is fixed independently of these examples.
Estimation	Bounded loss; sufficient sample size; reported uncertainty.

Good test performance supports good performance on future samples from the same S .

Finite-sample evidence, not correct behavior on every future input.

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Generalization: behavior and evidence

Low population error and a small training-to-population gap are different properties.

One sufficient learning argument:

$$\hat{R}_{\text{train}}(\hat{\theta}) \leq r, \quad \sup_{\theta \in \Theta} |R_S(\theta) - \hat{R}_{\text{train}}(\theta)| \leq \epsilon$$

$$\implies R_S(\hat{\theta}) \leq r + \epsilon$$

The uniform bound needs justification; any confidence qualification carries through.

A predictor can perform well without anyone collecting an independent test set.

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When does model error measure harm?

For error to correspond to L_F per order:

Population and labels	Ordering episodes match S ; labels give the intended binary choice.
Baseline execution	The prediction is faithfully ordered and charged before effective correction.

$$\Pr(L_F \text{ per order}) = R_S(\hat{\theta}) \quad \text{under those assumptions}$$

For zero-one loss to represent disutility:

$$u = a - b\ell_{01}, \quad b > 0 \quad \text{with fixed } a, b$$

Equal-cost errors, up to positive affine rescaling. Equal prices alone do not imply this utility shape.

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STPA 4: from a freshness shortcut to a charge

A hypothetical harm scenario

- 1 Development comparisons confound food type with freshness.
- 2 Both substitutes are fresh; the customer would choose spinach. The freshness-based predictor ties and returns $\hat{y} = -1$ for $x = \text{Apple}$.
- 3 The ordering rule submits the apple order: an unsafe control action occurs.
- 4 The merchant commits before notice. Later correction cannot prevent L_F .

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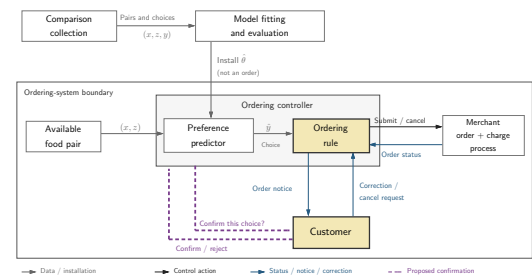
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From the scenario to requirements and evidence

Proposed requirement	Evidence to seek
Limit prediction error under the given ordering conditions to a justified tolerance.	Independent labeled comparisons under those conditions, with evaluation uncertainty reported.
Require explicit customer confirmation; without confirmation, do not allow nonrefundable commitment.	Log confirmed, rejected, delayed, and missing responses; inspect actual order and charge records.

Model-level requirement versus proposed system control; neither evidence substitutes for the other.

Proposed confirmation changes the control structure



Proposed, not implemented: without confirmation, do not allow nonrefundable commitment.

Roadmap

Learn a predictor inside an ordering system

Evaluate a claim, not just a score

Probe what the pipeline leaves unresolved

Good performance can leave a set

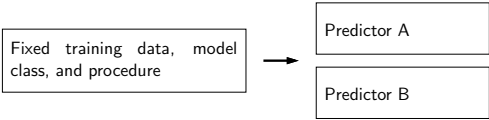
Good predictive performance may identify a set of acceptable predictors rather than one particular predictor.

$$\Theta_\epsilon^* = \{\theta \in \Theta : R_S(\theta) \leq \inf_{\theta' \in \Theta} R_S(\theta') + \epsilon\}$$

Which predictors can the pipeline return with the same effective validation performance?

The population-level set need not be the set of outputs of a particular pipeline.

Underspecification is a pipeline property



Otherwise incidental choices can yield distinct, validation-equivalent predictors.

This becomes consequential when those predictors differ in application-relevant behavior.

D'Amour et al., JMLR 2022

Three ways to stress-test a behavioral requirement

Test food-type preference at comparable freshness for the same customer.

- Stratified** Report performance on fresh-spoiled and matched-freshness subsets.
- Shifted** Collect a new population S' : both foods have comparable, acceptable freshness.
- Contrastive** For photographs, alter the background while preserving food and freshness; justify the expected invariance.

Each probes a specified behavioral requirement, not complete safety.

D'Amour et al., JMLR 2022

Test repeated runs, not just one predictor

- 1 Hold training data, model class, and procedure fixed.
- 2 Vary an otherwise incidental choice, such as a training seed.
- 3 Identify effectively validation-equivalent outputs, accounting for uncertainty.
- 4 Compare those outputs on the same application-relevant stress test.

Seed changes need not yield different predictors; one failed predictor is insufficient.

Evaluation also creates selection pressure

An evaluation criterion does two jobs:

Evidence	Produces evidence about a fixed predictor.
Selection	Helps determine which predictor we get.

Once evaluation results are used for selection, the selected predictor is no longer independent of that evaluation data.

Fresh held-out data, or an additional argument accounting for selection.

Discussion setting: food preferences

A predictor learns from human choices between pairs of foods. The intended construct is preference between food types at comparable freshness. Development comparisons may instead confound food type with freshness.

Use this setting for the stress test and repeated-run comparison.

Discussion: generalization and underspecification

Use the food-preference task and the baseline food-ordering control structure. Keep L_F , H_F , and the unsafe ordering action fixed.

For questions 2-4, connect the evaluation to the predictor's role in the harm scenario: state what observation your proposed evaluation would supply and what would remain unestablished about SC_F .

Do not assume that the proposed confirmation control has already been implemented.

Discussion: generalization

Come up with sufficient conditions for a predictor to generalize, how is this distinct from necessary conditions?

Discussion: stress testing

Design a stratified, shifted, or contrastive evaluation for the food-preference predictor.

Discussion: underspecification

What comparison across repeated training runs would demonstrate consequential underspecification, rather than merely one poorly performing predictor? Explain what consequential underspecification your group's stress test could elicit.

Discussion: selection pressure

If developers use those evaluation results to choose a predictor, what additional evidence is needed afterward?

Generalization and safety are different claims

Training	Fit to observed examples under the chosen loss.
Generalization	Behavior under a specified population distribution.
Stress testing	Behavior under a specified subset, shift, or intervention.
System safety	A harm-related claim requiring controls, evidence, and an adequate risk criterion.

Evaluation provides evidence and, when used for selection, helps determine which predictor we get.